Assessment on LAB-01

Even. $Y = \overline{(A+B.C).D}$

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

Odd. $Y = \overline{(A.(B+C))+D}$

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

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Exp. : 02

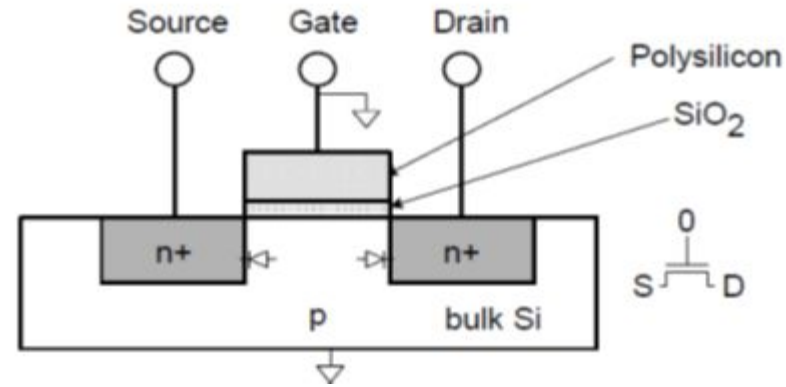
Design and Implementation of CMOS Logic Circuit Mask Layouts

Colour Codes :

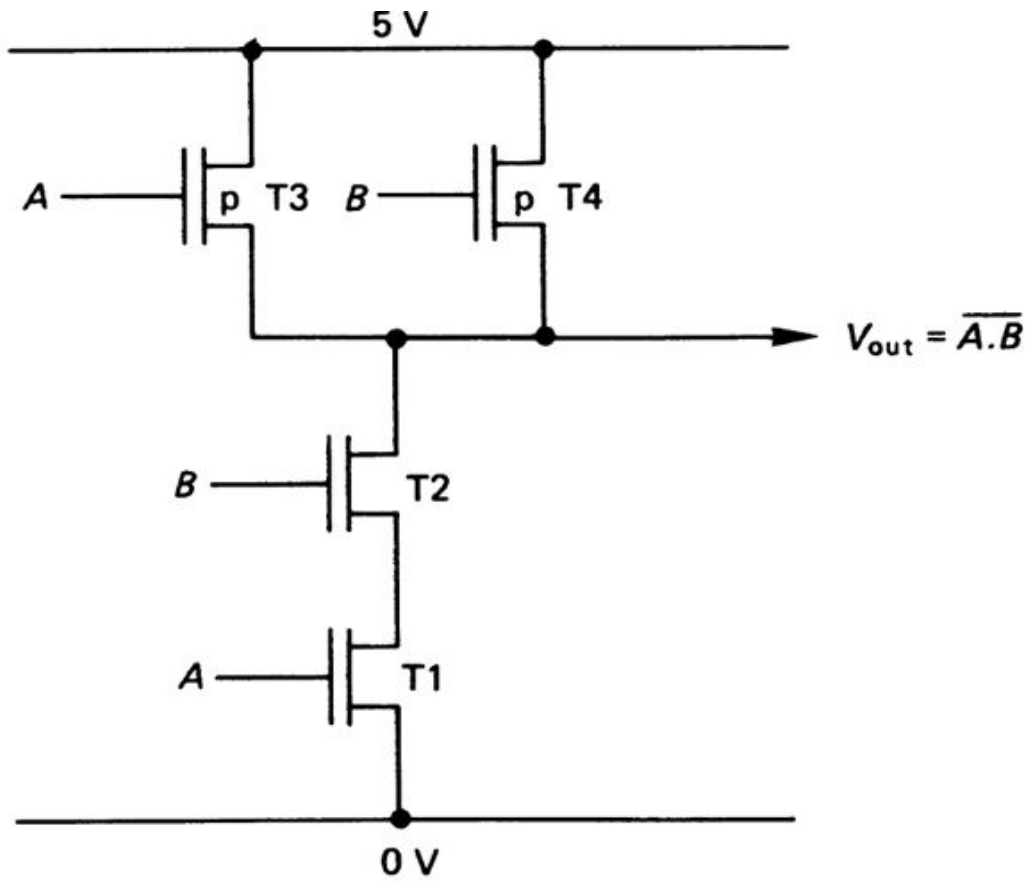
Layer	Color	Representation
N+ Active	Green	
P+ Active	Yellow	
PolySi	Red	
Metal 1	Blue	
Metal 2	Magenta	
Contact	Black	X
Buried contact	Brown	X
Via	Black	X
Implant	Dotted yellow	
N-Well	Dotted Green/Black	

Darshana Sankhe

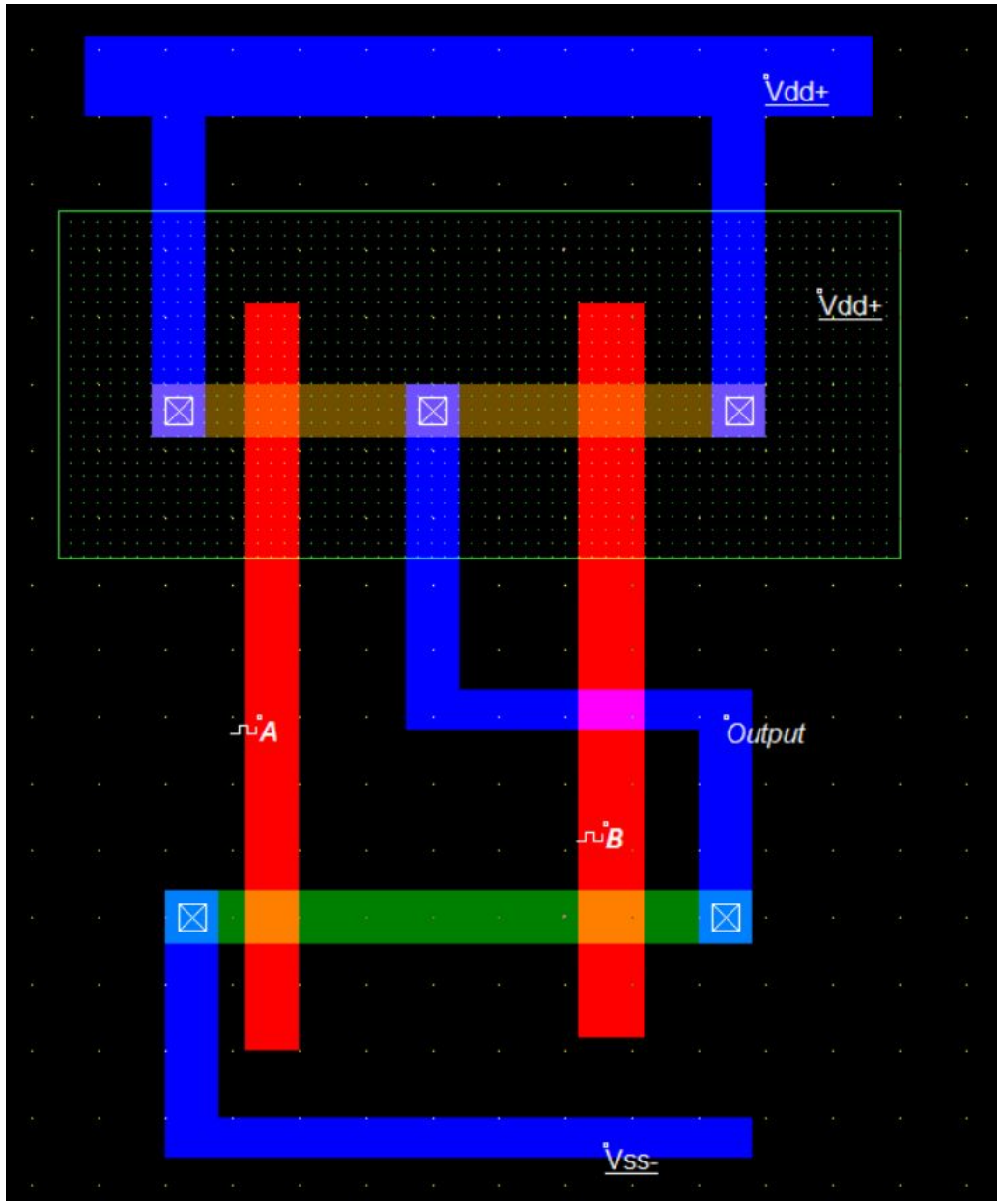
Enhancement-Mode MOSFET :



NAND

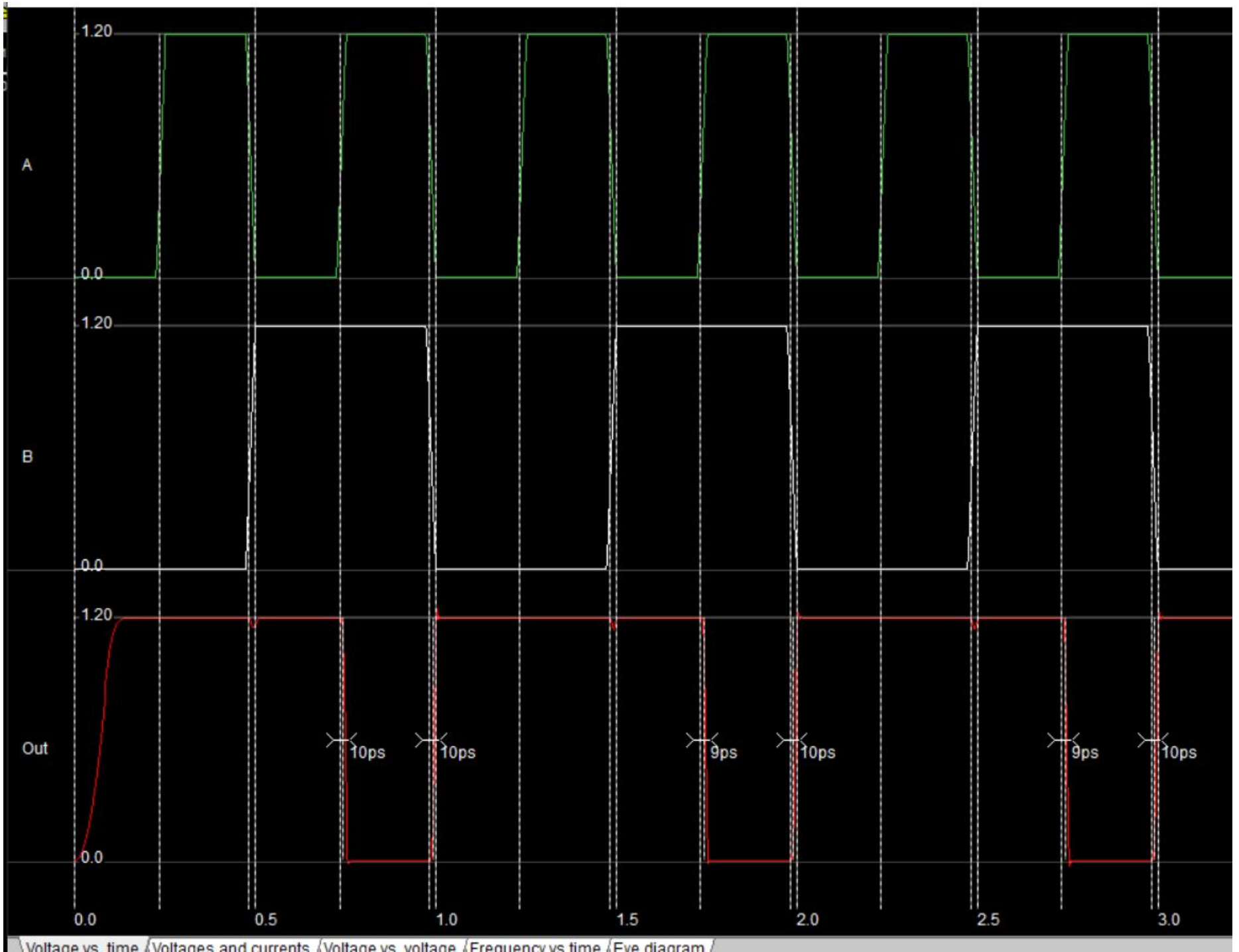


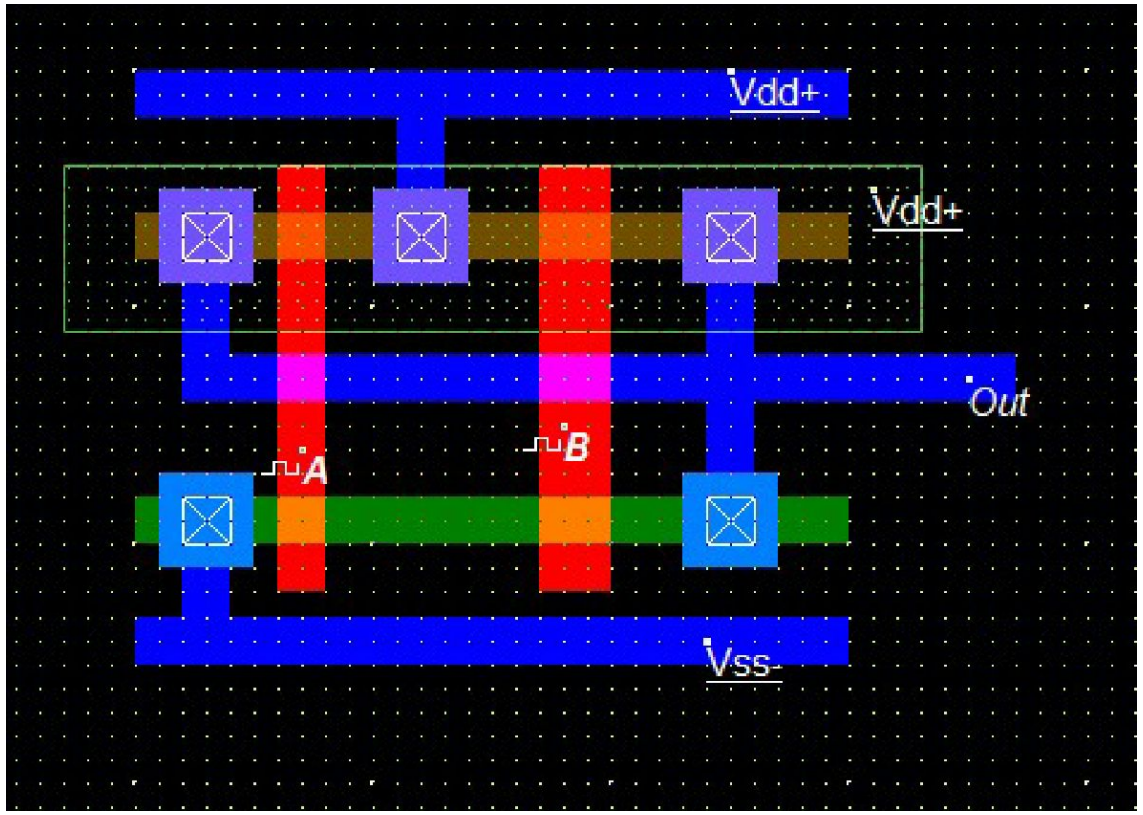
Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0



NAND

Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0




Vdd+
Vdd+
Out
A
B
Vss-

Problems:

1. $Y = \overline{A+B.C}$

A	B	C	Y=(A+B·C)'
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

2. $Y = \overline{(A+B).(C+D)}$

A	B	C	D	Y=((A+B)(C+D))'
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0